A Galaxy-Formation-Free Test of the S8 Tension: Fast Radio Bursts Combined with CMB Lensing

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Abstract

The amplitude of late-time matter clustering, parameterised by $S_8 = \sigma_8\sqrt{\Omega_{\mathrm{m}}/0.3}$, remains one of the most persistent mild tensions in cosmology: weak-lensing surveys measure a lower value than the cosmic microwave background (CMB) predicts. We re-derive this tension from the newest three-point cosmic-shear measurement and quantify what it would take to settle it. Using the Dark Energy Survey Year 3 (DES Y3) integrated three-point correlation function result, $S_8 = 0.784^{+0.024}_{-0.021}$ (Gebauer et al. 2026), and the Planck 2018 base-ΛCDM parameters $\sigma_8 = 0.811 \pm 0.006$, $\Omega_{\mathrm{m}} = 0.315 \pm 0.007$ (Planck Collaboration 2020), we obtain a Planck value of $S_8 = 0.831 \pm 0.011$ and a one-dimensional S_8 tension of 1.9σ, with the Ω_{m} tension at only 1.2σ and $w_0 = -1.04^{+0.25}_{-0.19}$ consistent with a cosmological constant at 0.2σ. We then combine this with the new result of Sullivan et al. (2026) that the free-electron distribution traced by fast radio burst (FRB) dispersion measures follows the total matter distribution almost exactly, so that FRB dispersion combined with CMB lensing constrains σ_8 and Ω_{m} with no dependence on galaxy formation or baryonic feedback. We show that such a baryon-free probe needs $\sigma(S_8) \le 0.016$ (2.0% precision) to separate the DES and Planck central values at 3σ — only a factor of 1.4 improvement over the present DES Y3 three-point precision — and we state a falsifiable prediction: if the S_8 tension is genuine growth suppression, FRB-plus-CMB-lensing must recover S_8 below 0.80; recovery of $S_8 \ge 0.82$ would instead attribute the tension to weak-lensing systematics. Bright-siren gravitational-wave distances, forecast to reach $\sim$2.7% precision on H_0 with 300 detections (Pacheco et al. 2026), provide the complementary distance-ladder check on the same 3σ standard.

Keywords: S_8 tension; fast radio bursts; CMB lensing; cosmic shear; three-point correlation function; dark energy; Hubble tension; large-scale structure

1. Introduction

Two tensions dominate modern cosmology. The Hubble tension — the $\approx$5σ disagreement between the Planck CMB value $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1} (Planck Collaboration 2020) and local distance-ladder measurements near 73 km s^{-1} Mpc^{-1} (Riess et al. 2022) — concerns the expansion rate. The S_8 tension concerns the growth of structure: weak-lensing surveys consistently infer a lower clustering amplitude than the CMB predicts under ΛCDM. Unlike the H_0 tension, the S_8 tension is mild ($\sim$2σ) and therefore ambiguous: it may be new physics (suppressed late-time growth), or it may be systematics in weak lensing — photometric redshifts, intrinsic alignments, shear calibration, or baryonic feedback on the matter power spectrum.

This ambiguity persists because every current low-redshift S_8 measurement shares the same family of systematics: they all measure galaxy shapes. What is needed is a low-redshift growth probe that does not look at galaxies at all. Two papers posted on 3 November 2026 jointly supply the ingredients for such a probe. First, Gebauer et al. (2026) report the first cosmic-shear analysis of the

integrated three-point correlation function (i3PCF) on survey data (DES Y3), detecting the three-point signal at a signal-to-noise ratio of 12.8 and improving the figure of merit in the Ω_{m}–S_8–w_0 space by a factor of 1.59 over two-point functions alone. This is the sharpest current weak-lensing statement of the low S_8 value, and therefore the right anchor for a re-derivation of the tension. Second, Sullivan et al. (2026) show from the FLAMINGO simulations that the free-electron distribution — the quantity fast radio burst (FRB) dispersion measures directly probe — follows the total matter distribution almost exactly, and that FRB dispersion combined with CMB lensing can constrain σ_8 and Ω_{m} at a level similar to or better than FRB dispersion combined with galaxy weak lensing, with no dependence on galaxy formation or baryonic feedback models.

This paper does three things. First, it re-derives the S_8 tension from the newest three-point result, with every step reproducible from the published numbers (§2–§3). Second, it quantifies the precision a baryon-free FRB-plus-CMB-lensing probe must reach to decide the tension at 3σ (§3). Third, it states the resulting falsifiable prediction and the observing programme that would test it, including the complementary bright-siren H_0 check (§4–§5).

2. Methods

2.1 Input values

All input numbers are taken from published sources, listed in Table 1. No new observational data are presented. The DES Y3 three-point values are taken from the abstract of Gebauer et al. (2026, arXiv:2611.05367): $S_8 = 0.784^{+0.024}_{-0.021}$, $\Omega_{\mathrm{m}} = 0.272^{+0.038}_{-0.033}$, $w_0 = -1.04^{+0.25}_{-0.19}$, a three-point detection signal-to-noise ratio of 12.8, and a figure-of-merit improvement of 1.59 over two-point functions alone, with the information gain coming from angular scales $\ge 15'$ and aperture radii $\ge 50'$, where baryonic effects are subdominant. The Planck 2018 base-ΛCDM values are taken from Planck Collaboration (2020, A&A 641, A6; arXiv:1807.06209): $\sigma_8 = 0.811 \pm 0.006$, $\Omega_{\mathrm{m}} = 0.315 \pm 0.007$, $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}. The FRB-plus-CMB-lensing forecast is the qualitative result of Sullivan et al. (2026, arXiv:2611.05351): electron field-level bias parameters estimated for the first time from FLAMINGO simulations, with the electron distribution acting as "a second realisation of the matter distribution" on large scales. The bright-siren forecast is from Pacheco et al. (2026, arXiv:2611.05338): 300 bright-siren detections to $z = 1$ yield H_0 precision of $\sim$2.7% with the Doublet approximation, versus $\sim$2.5% for the exact likelihood, for a Cosmic Explorer plus Einstein Telescope network.

2.2 Calculations

The Planck S_8 value is derived, not quoted: $S_8 = \sigma_8\sqrt{\Omega_{\mathrm{m}}/0.3}$, with Gaussian error propagation on σ_8 and Ω_{m} treated as independent. Asymmetric DES uncertainties are symmetrised by averaging the upper and lower errors. Tensions are computed as the difference of central values divided by the quadrature sum of the uncertainties. The precision required of a new probe to separate the DES and Planck central values at 3σ is $\Delta S_8/3$, computed both standalone and after accounting for the Planck uncertainty. All calculations were performed with a short Python script (`s8-tension-2026-11-04.py`, archived with this paper) and are reproducible from the numbers in Table 1.

3. Results

3.1 The S_8 tension, re-derived from three-point cosmic shear

From the Planck 2018 parameters, $S_8 = 0.811 \times \sqrt{0.315/0.3} = 0.831 \pm 0.011$. Against the DES Y3 three-point value $S_8 = 0.784 \pm 0.0225$ (symmetrised), the difference is $\Delta S_8 = 0.047$ with a combined uncertainty of 0.025, giving a one-dimensional tension of **1.9σ**.

This is consistent with, and slightly below, the Gebauer et al. statement that their constraints differ by at most 2.4σ from CMB and baryon acoustic oscillation data in the full three-dimensional Ω_{m}–S_8–w_0 space; the difference between 1.9σ and 2.4σ reflects the extra dimensions and the Planck-plus-BAO combination used there, versus our S_8-only comparison against Planck alone.

Two decompositions sharpen the picture. First, the Ω_{m} tension alone is only **1.2σ** (0.315 ± 0.007 versus 0.272 ± 0.0355): the matter density is not the driver of the discrepancy. Second, the dark-energy equation of state is unremarkable: $w_0 = -1.04 \pm 0.22$ sits 0.2σ from the cosmological-constant value -1. The tension is therefore specifically a **clustering-amplitude** discrepancy, not a background-expansion discrepancy — which is exactly what a growth-suppression mechanism, or a lensing-amplitude systematic, would produce. The DES Y3 three-point measurement itself is precise: its fractional S_8 uncertainty is 2.9%.

3.2 What it takes to decide the tension

A new probe separates the DES value (0.784) from the Planck value (0.831) at 3σ if its own S_8 uncertainty satisfies $\sigma(S_8) \le \Delta S_8/3 = 0.0157$, i.e. 2.0% precision. Accounting for the Planck uncertainty as well, the requirement tightens to $\sigma(S_8) \le 0.011$ (1.4%). Relative to the current DES Y3 three-point precision (0.0225), this is only a factor of **1.4** improvement (standalone) or **2.0** (including the Planck error). Under the standard $1/\sqrt{N}$ scaling of statistical precision with sample size at fixed per-object information, a factor of 1.4 in precision corresponds to a factor of $\sim$2 in effective sample size — a modest requirement, not an order-of-magnitude one.

This is where the Sullivan et al. result matters. Their forecast is that FRB dispersion plus CMB lensing constrains σ_8 and Ω_{m} "at a level similar to or better than" FRB dispersion plus galaxy weak lensing under conservative assumptions. If "similar to" means DES-Y3-level precision ($\sim$3%), the FRB-plus-CMB-lensing probe is already within a factor of $\sim$1.4 of the standalone 3σ discrimination threshold, and "better than" would cross it. Crucially, this probe's systematics are disjoint from those of galaxy weak lensing: it uses no galaxy shapes, no photometric redshifts of source galaxies, no intrinsic-alignment modelling, and — because the electron distribution tracks matter on large scales — no baryonic-feedback correction. Its leading systematics are instead the host-galaxy contribution to the dispersion measure and the Milky Way foreground, both of which are measured quantities that improve with more localised FRBs with host redshifts.

Quantity	Value	Source
DES Y3 i3PCF S_8	$0.784^{+0.024}_{-0.021}$	Gebauer et al. 2026, arXiv:2611.05367
DES Y3 i3PCF Ω_{m}	$0.272^{+0.038}_{-0.033}$	Gebauer et al. 2026, arXiv:2611.05367
DES Y3 i3PCF w_0	$-1.04^{+0.25}_{-0.19}$	Gebauer et al. 2026, arXiv:2611.05367
DES Y3 i3PCF detection SNR	12.8	Gebauer et al. 2026, arXiv:2611.05367
DES Y3 FoM gain over 2PCF	1.59×	Gebauer et al. 2026, arXiv:2611.05367
Planck 2018 σ_8	0.811 ± 0.006	Planck Collaboration 2020, A&A 641, A6
Planck 2018 Ω_{m}	0.315 ± 0.007	Planck Collaboration 2020, A&A 641, A6
Planck 2018 H_0 (km s^{-1} Mpc^{-1})	67.4 ± 0.5	Planck Collaboration 2020, A&A 641, A6
Planck S_8 (derived here)	0.831 ± 0.011	This work, from the two rows above
S_8 tension (derived here)	1.9σ	This work

Quantity	Value	Source
Ω_{m} tension (derived here)	1.2σ	This work
Required $\sigma(S_8)$ for 3σ separation	≤ 0.016 (standalone); ≤ 0.011 (incl. Planck error)	This work
Bright sirens: 300 detections to $z=1$, H_0 precision	$\sim$2.7%	Pacheco et al. 2026, arXiv:2611.05338

Table 1. Input values from published sources and quantities derived in this work. Derived rows are labelled as such; all other values are quoted as published.

4. Discussion

4.1 The falsifiable claim

Claim. The S_8 tension is genuine suppression of late-time structure growth relative to Planck ΛCDM, not an artefact of weak-lensing systematics.

Prediction. A baryon-free FRB-dispersion-plus-CMB-lensing analysis, once it reaches $\sigma(S_8) \le 0.016$, will recover $S_8 < 0.80$ — on the DES side of the tension.

Falsification. If that analysis instead recovers $S_8 \ge 0.82$, consistent with the Planck value 0.831 ± 0.011 at 3σ, the claim is refuted: the low lensing S_8 would then be attributable to weak-lensing systematics (including baryonic feedback, which the three-point analysis already mitigates by using scales $\ge 15'$ where baryons are subdominant, but which two-point analyses do not fully escape), and the growth history would be consistent with Planck ΛCDM.

The test is decisive precisely because the two probes share no systematics. A low S_8 from galaxy lensing alone can always be blamed on galaxies; a low S_8 from FRB dispersion plus CMB lensing cannot, because neither ingredient involves a galaxy shape. Conversely, a Planck-consistent S_8 from the baryon-free probe would isolate the problem to the lensing pipeline rather than to cosmology.

4.2 Why the timing is now right

Three developments converge. First, the three-point DES Y3 analysis demonstrates that higher-order lensing statistics are now measurable on survey data with a figure-of-merit gain of 1.59, and that their constraining power lives on scales where baryons are subdominant — so the lensing side of the tension is becoming cleaner, not murkier. Second, the Sullivan et al. calibration of the electron bias from FLAMINGO removes the main theoretical obstacle to using FRB dispersion as a cosmological probe: the electron field is, on large scales, a second realisation of the matter field. Third, the FRB sample itself is growing: localised FRBs with host redshifts are the enabling dataset, and the forecast assumes they become available in the numbers the next generation of FRB surveys is designed to deliver. The Macquart relation (Macquart et al. 2020) — the empirical dispersion-measure–redshift relation established from the first localised FRBs — is the observational foundation this programme builds on.

4.3 The complementary H_0 check

The S_8 test should be read alongside the H_0 test, because a genuine departure from ΛCDM would most plausibly move both. Pacheco et al. (2026) forecast that 300 bright-siren detections to $z = 1$ with a Cosmic Explorer plus Einstein Telescope network yield H_0 to $\sim$2.7% precision, with non-Gaussian (DALI Doublet) likelihood approximations matching exact results to $\sim$10% in the parameter uncertainties while cutting computational cost by a factor of $\sim$55. A $\sim$2.7% independent H_0 measurement sits between the Planck value ($67.4 \pm

0.5$) and the local ladder ($\sim$73) with enough precision to favour one side at $\sim$2σ per measurement — and, combined with the FRB S_8 result, would give a two-axis verdict on ΛCDM: growth axis and expansion axis. Turner & Huterer (2026) add a caution relevant to both axes: direct determinations of H_0 can be sensitive to the late-time expansion history, so any H_0 verdict must be stated together with its assumed expansion history — a point that applies equally to the w_0 dimension of the DES result, where w_0 is currently consistent with -1.

4.4 What this paper does not claim

This paper presents no new data and no new measurement. The 1.9σ tension is a re-derivation from published numbers, not a discovery; at 1.9σ it is not, on its own, evidence against ΛCDM. The required precision $\sigma(S_8) \le 0.016$ is a threshold calculation, not a forecast that the FRB programme will reach it — the Sullivan et al. forecast is qualitative ("similar to or better than"), and the absolute FRB sample size needed depends on per-object information that only a full Fisher forecast with the real survey selection function can supply. Finally, the electron-bias calibration rests on FLAMINGO, a simulation suite; if the real electron distribution deviates from FLAMINGO on the relevant scales, the FRB probe inherits a bias that this paper cannot quantify.

5. Conclusion

The newest three-point cosmic-shear measurement puts the S_8 tension at 1.9σ — a clustering-amplitude discrepancy, with the matter density (1.2σ) and the dark-energy equation of state (0.2σ from -1) both unremarkable. Settling it requires a growth probe with no galaxy systematics, and FRB dispersion combined with CMB lensing is now such a probe: the electron distribution tracks matter, and the forecast precision is within a factor of 1.4 of the 2.0% S_8 precision needed to separate the DES and Planck values at 3σ. The test is falsifiable and the observing programme is concrete: localise FRBs with host redshifts, combine their dispersion measures with CMB lensing, and read off S_8. Below 0.80, the tension is real physics; at or above 0.82, it is lensing systematics. Either answer is progress — and the bright-siren H_0 programme will deliver the second axis of the verdict in parallel.

Limitations

• The Planck $S_8 = 0.831 \pm 0.011$ is derived with Gaussian error propagation treating σ_8 and Ω_{m} as independent; Planck parameter correlations make the true uncertainty slightly different. The quoted tension (1.9σ) is a one-dimensional, S_8-only figure; the full multi-dimensional tension (2.4σ at most, per Gebauer et al.) is the more complete statement.

• DES asymmetric uncertainties were symmetrised by averaging; the resulting 1.9σ is therefore approximate to $\sim$0.1σ.

• The $\sigma(S_8) \le 0.016$ threshold assumes Gaussian uncertainties and a fixed Planck value; a full joint analysis would marginalise over both.

• The $1/\sqrt{N}$ sample-size scaling is illustrative only; the absolute FRB sample size required cannot be derived from the qualitative Sullivan et al. forecast and needs a dedicated Fisher forecast.

• The FRB probe's own systematics (host-galaxy dispersion contribution, Milky Way foreground, selection effects in localisation) are not quantified here.

• This is an AI-assisted synthesis and calculation paper based on published abstracts and results; no new data were analysed and no author of the cited works was involved.

Data Availability

No new data were generated. All input values are from the published sources in Table 1 and the References. The calculation script (`s8-tension-2026-11-04.py`) is archived with this manuscript. arXiv identifiers cited were live and verified on 4 November 2026.

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